VisionForge AI – Project 2

Analytics and Logging Report

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Introduction

VisionForge AI is an accessibility-focused web application designed to assist visually impaired users by analyzing images and providing structured, navigation-oriented descriptions.

This report focuses on the system's analytics and logging capabilities. It evaluates how user data is captured, stored, and interpreted, and how effectively the system performs under real usage conditions.

The purpose of this report is to analyze system behavior, identify trends, evaluate performance, and highlight limitations.

System Logging Overview

The system logs all major user interactions through backend API routes. Every time a user interacts with the assistant, relevant data is captured and stored.

The logged data includes:

- User query
- Image metadata
- AI-generated description
- Input type (image or voice)
- Timestamp
- Associated user

All logs are stored in Google Datastore under the entity VisionForgeAnalysisLog.

Datastore Logs Table

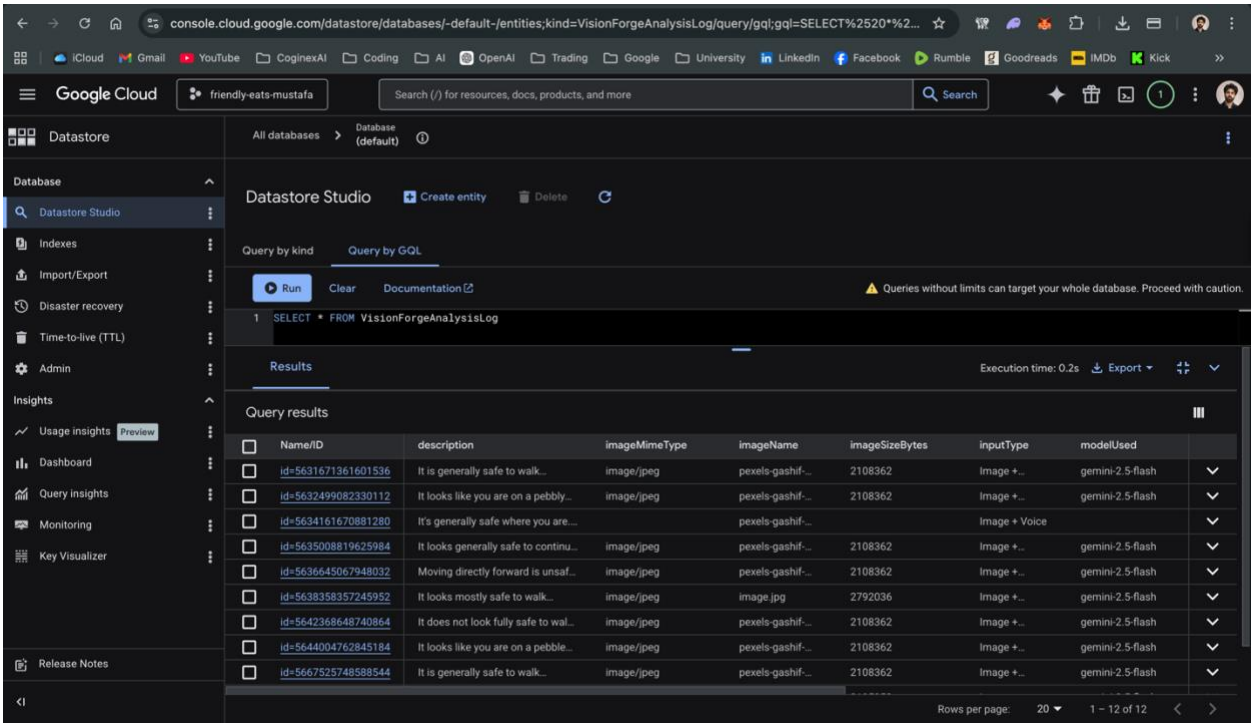

Figure 1: This screenshot shows stored logs including user queries, AI-generated descriptions, and timestamps.

API Logging and Request Flow

Each user interaction triggers backend API calls. These calls are essential for processing requests and are useful for debugging and monitoring system behavior.

Key API routes include:

- /api/analyze for image and question processing
- /api/logs for retrieving stored interaction data
- /api/auth/login for user authentication

Terminal Logs

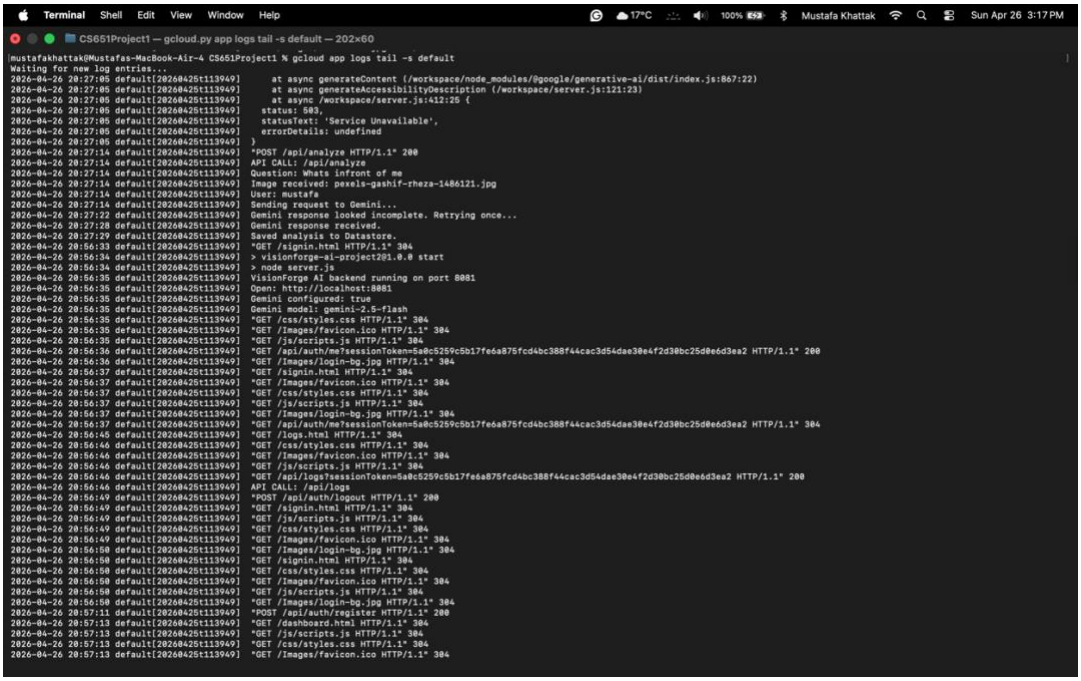

Figure 2: This screenshot shows real-time API calls triggered during user interaction.

Backend Code

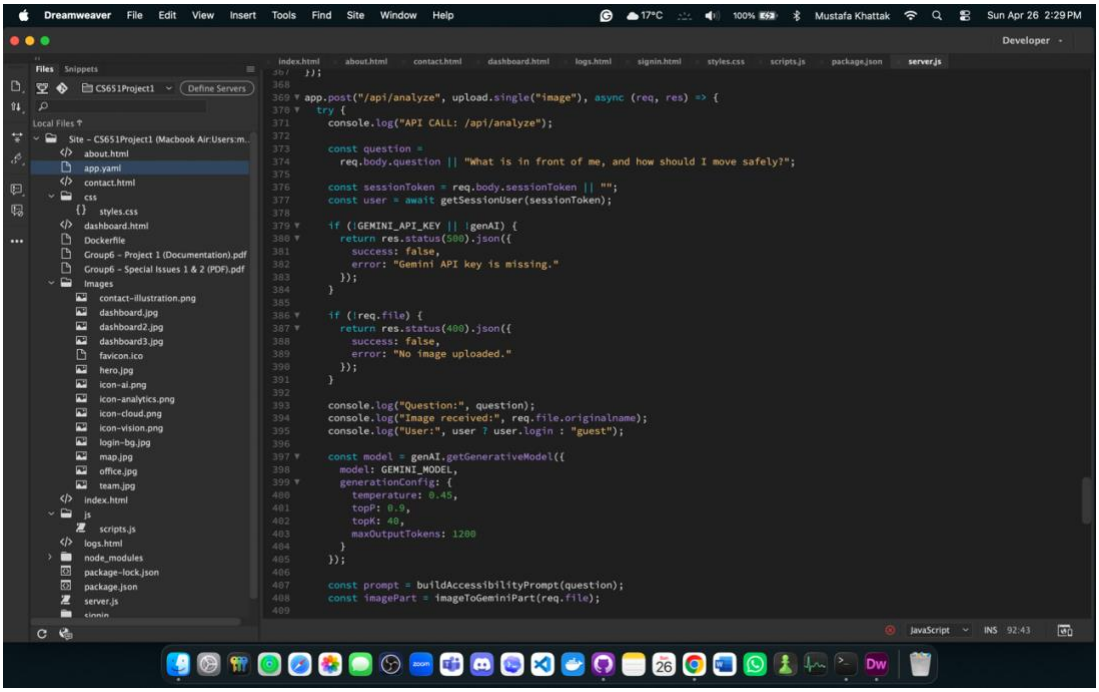

Figure 3: This code handles image processing, communication with Gemini AI, and saving results to the database.

Data Collection and Storage Behavior

The system automatically stores data after each successful interaction.

The process works as follows:

1. The user uploads an image or captures one using the camera
2. The user enters a question
3. The backend sends the data to Gemini AI
4. Gemini generates a structured response
5. The backend stores the response in the database
6. The logs become visible on the logs page

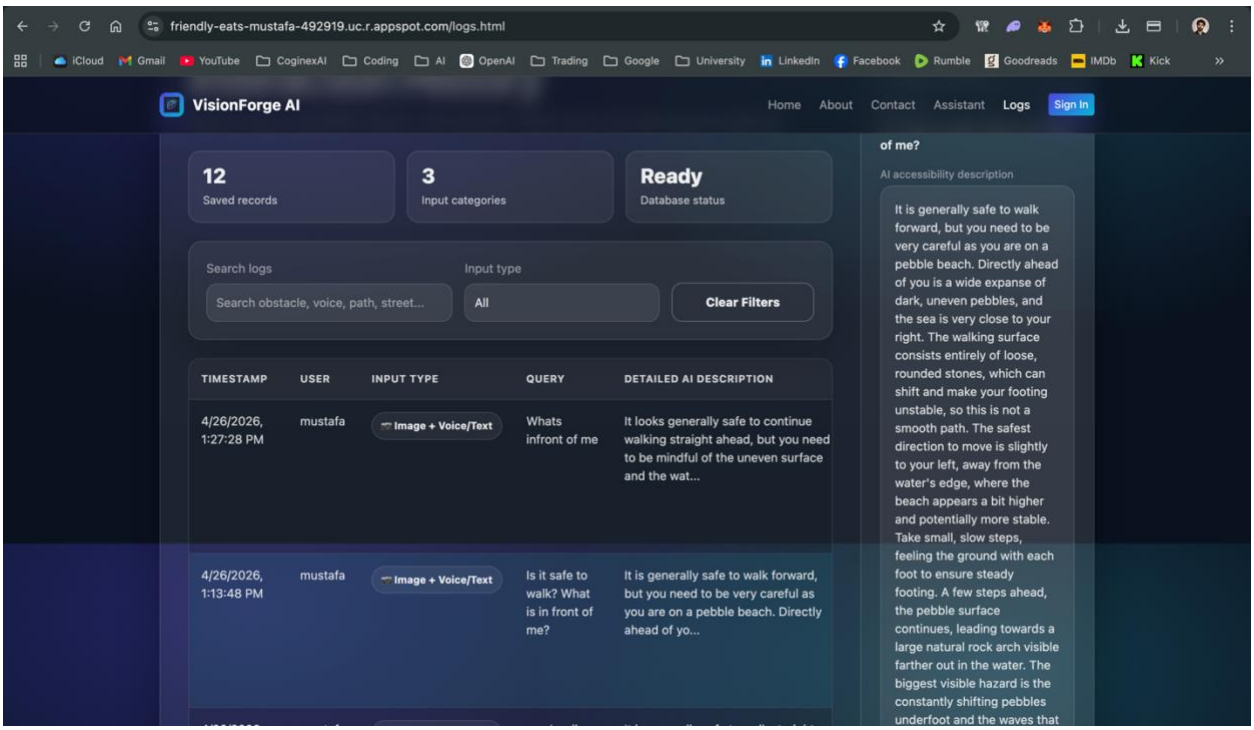

Figure 4: This screenshot shows how stored data is retrieved and displayed to the user.

Analytical Interpretation

Based on the collected logs, several patterns can be observed.

Most user queries are related to navigation and safety, such as identifying obstacles or determining whether a path is safe to walk on.

The AI responses are structured in a consistent format, typically including:

- A safety summary
- A description of the environment
- Navigation guidance

This structured output improves usability for visually impaired users.

Data Trends

Several trends are visible from system usage.

First, users often analyze the same image multiple times with different questions, indicating that they rely on the system for a deeper understanding rather than a single response.

Second, most queries are safety-focused. Users tend to ask about obstacles, walking direction, and surface conditions.

Third, the AI responses are consistent in structure, which improves readability and reliability.

Comparative Analysis

Compared to basic image captioning systems, VisionForge AI provides more practical and useful output.

Basic image captioning typically generates general descriptions of an image. In contrast, VisionForge AI focuses on accessibility by providing structured guidance.

It includes navigation advice and safety warnings, which are not present in standard captioning systems.

This makes the system more suitable for real-world use by visually impaired users.

Performance Analysis

The system performs well under normal usage conditions.

Strengths include:

- Fast response time for AI-generated descriptions
- Accurate recognition of environments and objects
- Consistent and structured output
- Smooth integration between frontend, backend, and database

However, some weaknesses were observed:

- Occasional incomplete responses in rare cases
- Performance depends on image quality
- Requires an internet connection and API availability

Analytics Limitations

There are several limitations in the current analytics implementation.

The dataset is relatively small, which limits the ability to perform deep statistical analysis.

The system does not track advanced metrics such as response time, error rates, or user behavior patterns.

There is no visual analytics dashboard, such as charts or graphs, to represent trends.

Additionally, errors are displayed to the user but are not stored for further analysis.

Identified Deficiencies

Some deficiencies in the system include:

- Lack of quantitative performance metrics
- No filtering or advanced search functionality in logs
- No visualization of collected data
- No export or reporting features
- Limited analytical depth due to a small dataset

Improvements and Future Work

Several improvements can be made to enhance the system.

These include:

- Adding an analytics dashboard with charts and trends
- Tracking response time and performance metrics
- Implementing filtering and sorting for logs
- Storing error logs for debugging and analysis

- Supporting video-based analysis
- Improving AI prompts for more detailed guidance

Conclusion

VisionForge AI successfully demonstrates a functional and practical system that integrates AI, web development, and cloud-based data storage.

The application provides real-time accessibility-focused analysis and maintains persistent logs of user interactions.

While the system performs well in its current form, there is room for improvement in analytics depth and performance tracking.

Overall, the project represents a strong implementation of a real-world AI application with meaningful impact.